



mesohyal®
intradermal
biostimulation:
from cell to skin

the element for skincare experts

Intradermal medical solutions

Benefits of discipline

Intradermal route or mesotherapy is a medical procedure that consists of localised, superficial micro-injections into the skin to correct multiple aesthetic manifestations, such as wrinkles, photoaging of the skin, dehydration, flaccidity, cellulitis and alopecia, among others.

Direct administration

The product is administered in the area to be treated and efficacy is increased by crossing the epidermal barrier.

Micro-dosing

Maximises absorption capacity and improves skin response.

Physical stimulation

The action of the needle triggers a physical reaction, stimulating tissue repair and regeneration.

Multiple indications

These solutions make it possible to treat facial, body and capillary blemishes.

Minimally invasive procedure

Does not require preparation or recovery and allows the patient to return to daily life immediately.

Combined modalities

A technique that is compatible with other common medical-aesthetic procedures, such as fillers, botulinum toxin, threading and chemical peels, thereby improving and maintaining results.

Recommended techniques

Intradermal injections can be performed manually or by specific devices that allow an assisted practice. The injection techniques require a specific prior training imparted by expert hands that ensures the correct administration and differentiation of each of the main recommended techniques:



Intradermal Nappage



Intradermal papule



Point-to-point



Tracing

Assisted technique: **mct injector**®

Electronic device that combines 2 complementary techniques.

• Intradermal

3 programmes the professional can use to modulate the administered dose and the depth and speed of the needle for greater precision and comfort for the specialist, with nappage, point-to-point and retro-tracing techniques.

mct injector is compatible with standard central cone syringes (not Luer-Lock) and with all types of needles.

• Carboxytherapy

CO₂ injection system that improves microcirculation, increases oxygenation and stimulates collagen synthesis.



mesohyal™ key features



Proven efficacy

The **mesohyal** range has been tested in vitro and in vivo to guarantee the maximum therapeutic response and dermal tolerance.



Sterile devices

Packaging that preserves the quality and efficacy of the active ingredients over time.



Free hyaluronic acid base

All **mesohyal** products are formulated with high-quality free hyaluronic acid of biotechnological origin.



Cosmeceutical complementarity

A wide range of products that allow their combination to compose personalized treatments, adapted to the needs of each patient.



Selected active ingredients

Active ingredients with specific targets that treat the cell precisely by acting on biological skin processes.



CE marking

Symbol that certifies and endorses the quality and safety guarantee system.⁽¹⁾



Excellence

Rigorous production process with the highest quality and safety standards.

Intradermal medical solutions

mesohyal™: a full range for a customised solution

Cellular treatment with biomimetic peptides

Peptides act as **regulators of biochemical cellular mechanisms**, through the interaction of their specific amino acid sequence with receptors or endogenous molecules.

The synthesis of analogue sequences can fulfil the same functions regulating cellular processes in the body.

mesoestetic® presents two new **formulations with biomimetic peptides**, compounds of specific sequences that act activating dermal regeneration processes and accelerate the mechanisms that **reverse cellular aging**.

A new generation of highly effective, highly specific biological active ingredients.



5 vials x 3 ml

mesohyal™ argibenone

Reduction of dynamic expression lines and photoinduced damage⁽²⁾.

Antiaging, antioxidant solution that combines a biomimetic hexapeptide, idebenone and hyaluronic acid.

[Free HA]: 15mg/ml



5 vials x 3 ml

mesohyal™ redenx

Reduces gravitational wrinkles, enhances density and firmness⁽³⁾.

Global antiaging solution that combines a tripeptide and a tetrapeptide to activate the synthesis of ECM* proteins*.

[Free HA]: 15mg/ml



5 vials x 3 ml

mesohyal™ hyaluronic

Regeneration⁽⁴⁾, optimisation of cell functions, rehydration.

Solution with a high concentration of non-reticulated hyaluronic acid.

[Free HA]: 15mg/ml



5 vials x 5 ml

mesohyal™ NCTC 109

Intensive cell biorevitalisation.

Multivitamin solution with amino acids, mineral salts, coenzymes and hyaluronic acid.

[Free HA]: 0,025mg/ml



5 vials x 3 ml

mesohyal™ X-DNA

Protection of the cell membrane, tissue repair⁽⁵⁾.

Highly polymerised sodium polynucleotides in a hyaluronic acid solution.

[Free HA]: 2,5mg/ml



10 vials x 5 ml

mesohyal™ organic silicon

Tissue restructuring⁽⁶⁾ and cell activator.

Solution rich in essential organic silicon for optimum organisation of the ECM* in a hyaluronic acid base.

[Free HA]: 2,5mg/ml



20 ampoules x 5 ml

mesohyal™ vitamin C

Luminosity, tone uniformity⁽⁷⁾, antioxidant⁽⁸⁾.

Hyaluronic acid solution with ascorbic acid (20%) and mineral salts.

[Free HA]: 2,5mg/ml



20 ampoules x 5 ml

mesohyal™ DMAE

Skin tensor, lifting effect.

Hyaluronic acid solution with a high concentration of dimethylaminoethanol (DMAE).

[Free HA]: 2,5mg/ml



20 ampoules x 5 ml

mesohyal™ oligoelements

Stimulation of the extracellular matrix.

Hyaluronic acid solution with a combination of 9 oligoelements (Zn, Mn, Mb, Ni, Co, Se, Cu, Mg, Vn).

[Free HA]: 2,5mg/ml



20 ampoules x 5 ml

mesohyal™ carnitine

Lipolytic, favours fat metabolism⁽⁹⁾.

High concentration of L-carnitine in a hyaluronic acid base.

[Free HA]: 2,5mg/ml



20 ampoules x 5 ml

mesohyal™ artichoke

Draining, anti-cellulite⁽¹⁰⁾, detoxifying.

Artichoke extract in a hyaluronic acid base.

[Free HA]: 2,5mg/ml



20 ampoules x 2 ml

mesohyal™ melilot

Favours microcirculation.

Hyaluronic acid solution with sweet clover extract, troxerutin and mineral salts.

[Free HA]: 2,5mg/ml



20 ampoules x 2 ml

mesohyal™ biotin

Reactivation of the capillary metabolism.

Hyaluronic acid solution with biotin, B vitamins and mineral salts.

[Free HA]: 2,5mg/ml

*ECM: extracellular matrix

Intradermal medical solutions

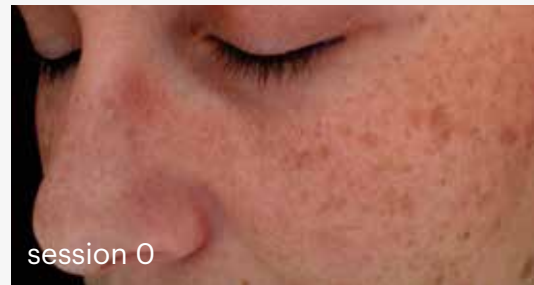
mesohyal™ facial: clinical cases

Clinical study protocol⁽¹¹⁾ “skin recharge C”:

1 mesohyal™ NCTC 109 + 1/2 mesohyal™ vitamin c

X5 sessions

Gender: female **Age:** 39 **Phototype:** III
Improved luminosity and reduced hyperpigmentation in the malar area⁽¹¹⁾.

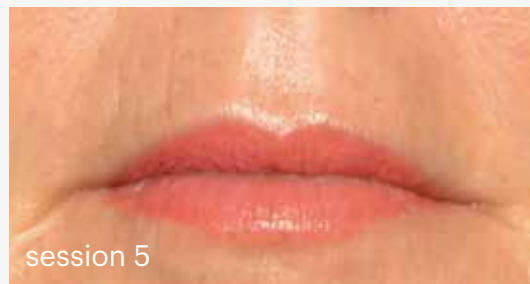
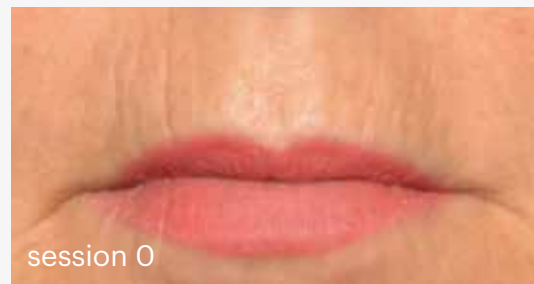


Clinical study protocol⁽¹²⁾ “age skin repair”:

1 mesohyal™ hyaluronic + 1/2 mesohyal™ organic silicon

X5 sessions

Gender: female **Age:** 47
Improvement of ‘bar codes’⁽¹²⁾.



Gender: female **Age:** 46
Reduction of cutaneous expression lines in the forehead area⁽¹²⁾.



mesohyal™ facial: clinical cases

Clinical study protocol⁽¹³⁾ “skin firm & lift”:

1 mesohyal™ X-DNA + 1/2 mesohyal™ DMAE

X5 sessions

Gender: female **Age:** 49
Improved firmness in the periauricular and malar area⁽¹³⁾.

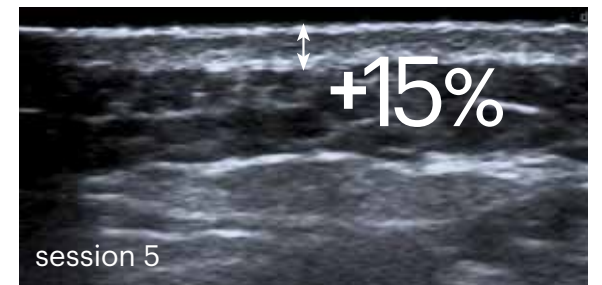
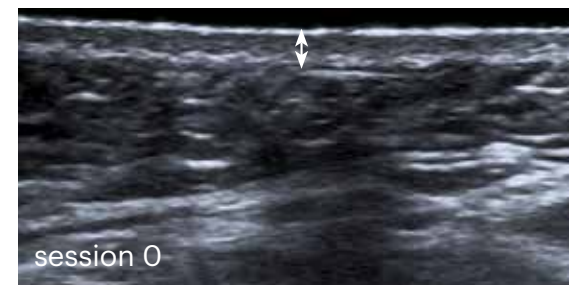


Gender: female **Age:** 49
Reduction of external perioral lines⁽¹³⁾.



Ultrasound measurement of the cutaneous tissue⁽¹⁴⁾:

Significant increase in dermal thickness (+15%).



Intradermal medical solutions

mesohyal™ facial: clinical cases

Clinical study⁽²⁾ mesohyal argibenone:
correction of dynamic wrinkles and photoaging.

1 mesohyal™ argibenone X5 sessions

Gender: female **Age:** 49
Correction of expression lines in the periauricular area⁽²⁾.




Gender: female **Age:** 43
Reduction of cutaneous expression lines in the forehead area⁽²⁾.




Clinical study⁽³⁾ mesohyal redenx:
Reduction of gravitational wrinkles and increased firmness.

1 mesohyal™ redenx X5 sessions

Gender: female **Age:** 52
Redensifying and increase in the firmness of the lower third of the face⁽³⁾.




Gender: female **Age:** 47
Redensifying and increase in the firmness of the lower third of the face⁽³⁾.



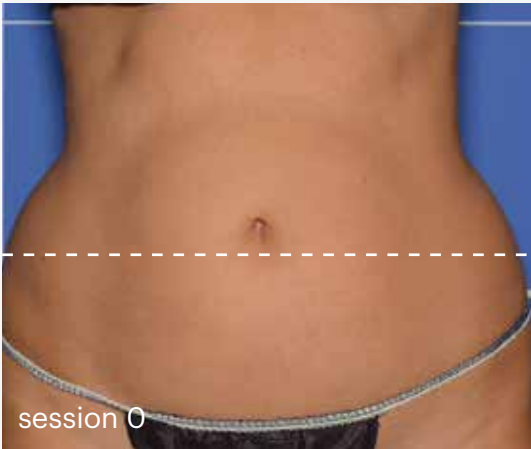
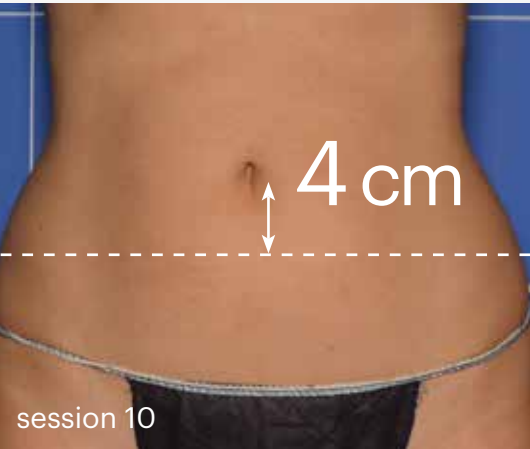

mesohyal™ corporal: clinical cases

Clinical study protocol⁽¹⁵⁾ lipolítico:

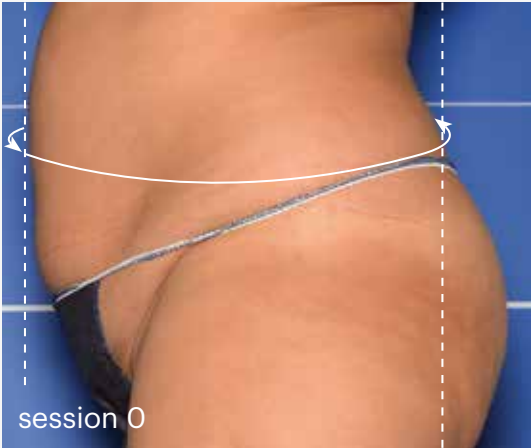
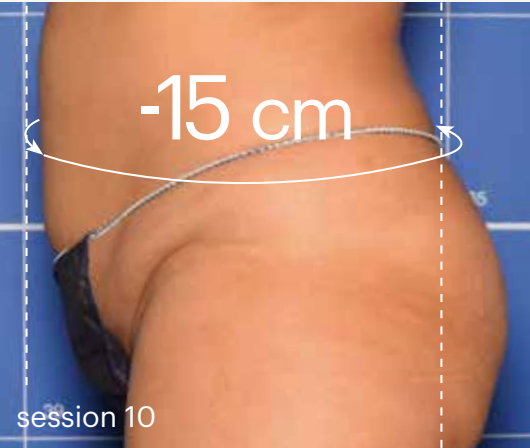
PHASE 1: 1 mesohyal™ melilot + 1 mesohyal™ carnitine + 1 mesohyal™ organic silicon X5 sessions

PHASE 2: 1 mesohyal™ melilot + 1 mesohyal™ organic silicon + 1 mesohyal™ DMAE X5 sessions

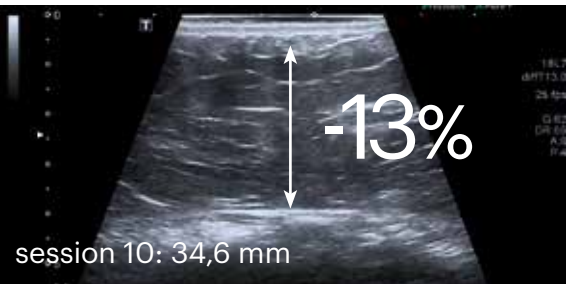
Gender: female **Age:** 42
Following the lipolytic protocol, an elevation of the tissue and a significant reduction in the abdominal contour of 10 cm, particularly visible in the flank areas, was observed⁽¹⁵⁾.

Gender: female **Age:** 36
Reduction of the abdominal contour of 15 cm due to the intensive drainage and lipolytic action of the treatment administered⁽¹⁵⁾.

Ultrasound measurement of the adipose tissue⁽¹⁶⁾:

Intradermal medical solutions

mesohyal™ capilar: clinical cases

Clinical study protocol⁽¹⁷⁾ "trichology":

1/2 mesohyal™ NCTC-109 + 1 mesohyal™ biotin

X7 sesiones

Gender: male **Age:** 36

Generalised increase in the density and diameter of the hair in the entire scalp area⁽¹⁷⁾.



Gender: male **Age:** 43

Significant increase in hair density and diameter in the frontal-parietal zone⁽¹⁷⁾.



mesohyal™: post procedure gel cream

INTRADERMAL MEDICAL SOLUTIONS

Cosmeceutical home treatment specifically developed for enhancing and prolonging the antiaging effects of facial treatments with mesohyal.

formula and properties



Polynucleotides

Cell DNA fractions to stimulate cell repair and regeneration and enhance collagen synthesis.



Pro-hyaluron system®

Innovative complex based on hyaluronic acid and antihyaluronidase active ingredient to prevent HA degradation.



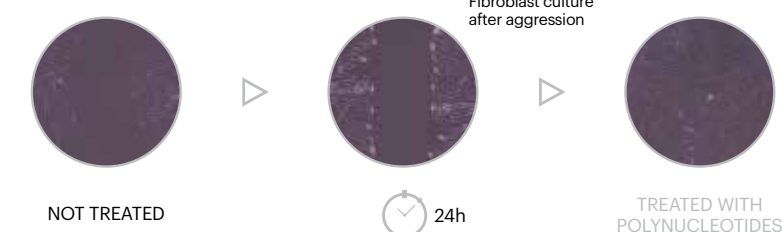
Pollutox®

A plant complex and a polymer provide a protective action against contaminating agents.

efficacy studies

Cell regeneration⁽¹⁸⁾

Fibroblast repairing activity after an aggression.
Scratch assay



Collagen synthesis⁽¹⁹⁾

≈ x2

Increase of collagen synthesis measured in human fibroblasts cultured in polynucleotides and hyaluronic acid mesohyal after 48h vs the control. *Elisa method*

Inhibition of HA degradation⁽²⁰⁾

≈ -15%

Inhibition of HYAL (hyaluronidase coding gene) measured in human fibroblasts cultured in Pro-hyaluron system® after 24 h vs control. *Measures obtained with qPCR*

at home

How to use:

Apply over the intact skin a dose of about 2-3 ml over the face, neck and décolleté. For an intensive action it is recommended to **apply it day and night.**

Precautions:

Topical use. Do not apply over wounds and/or in case of intense skin sensitisation.

Intradermal medical solutions

The most complete and extensive range of healthcare products for intradermal administration.

CE marking: guarantee of safety and efficacy.

Versatile treatment designed for individual application or administration of **personalised applications** to cover multiple facial, body and capillary indications.

Carefully selected ingredients that act in different cell targets to **repair and treat** tissues precisely.

Exclusive formula with a **hyaluronic acid base** and selected active ingredients of recognised therapeutic efficacy.

Compatible **with other medical procedures** (botulinum toxin, fillers, peeling, laser, pulsed light, etc.).



1. Intradermal medical devices are controlled by independent official agencies that certify their quality through CE marking. **2.** Dr. Tatyana Bayateva; Dr. Luis Shotze. An Open-Label Uncontrolled Single Center Study for the Evaluation of the Efficacy and Safety of the Dermal Filler "mARG-01-17" in the facial rejuvenation therapy. 2019. **3.** Dr. Șerban Roșu; Dr. Luis Shotze. A multicentre, open-label, interventional, pilot trial to assess the performance and tolerance of mRDX-02-17 dermal filler in the facial rejuvenation. 2019. **4.** Unit of Biotechnology. Study on the capacity of dermal regeneration (fibroblast proliferation and migration) of mesohyal hyaluronic. mesoestetic Pharma Group S.L. 2016. **5.** Unit of Biotechnology. Study on the collagen synthesis capacity of mesohyal X-DNA. mesoestetic Pharma Group S.L. 2016. **6.** Unit of Biotechnology. Study in vitro on the efficacy of organic silicon on collagen synthesis. mesoestetic Pharma Group, S.L. 2017. **7.** Unit of Biotechnology. Study on melanin synthesis for mesohyal Vitamin C. mesoestetic Pharma Group S.L. 2015. **8.** Unit of Biotechnology. Study on the antioxidant capacity against DPPH radical for mesohyal Vitamin C. mesoestetic Pharma Group, S.L. 2015. **9.** Unit of Biotechnology. Study in vitro on the efficacy of adipogenesis and lipolysis of mesohyal Carnitine. mesoestetic Pharma Group, S.L. 2015. **10.** Unit of Biotechnology. Study in vitro on the lipolysis stimulating effect of mesohyal Artichoke. mesoestetic Pharma Group, S.L. 2015. **11.** Medical Unit. Study in vivo on the depigmenting action of mesohyal Vitamin C and mesohyal NCTC 109. mesoestetic Pharma Group, S.L. 2016. **12.** Galcerán F, Lorente E, Vidal R. Study on skin elasticity and hydration after intradermal administration of mesohyal HYALURONIC – mesohyal ORGANIC SILICON. mesoestetic Pharma Group S.L. 2016. **13.** Galcerán F, Lorente E, Vidal R. Study on skin firmness and hydration after intradermal administration of mesohyal X-DNA - DMAE. mesoestetic Pharma Group S.L. 2016. **14.** Vidal R. Ultrasonic assessment of skin tissue after administration of mesotherapy (mesohyal™). mesoestetic Pharma Group S.L. 2016. **15.** Galcerán F, Lorente E, Vidal R, Luis L. Study on the draining, lipolytic and firming effect after administration of mesotherapy (mesohyal™) in the body. mesoestetic Pharma Group S.L. 2017. **16.** Vidal R. Ultrasonic evaluation of fat tissue (subcutaneous plane) after administration of mesotherapy (mesohyal™). mesoestetic Pharma Group S.L. 2017. **17.** Galcerán F, Lorente E, Vidal R, et al. Study on the strengthening and increase of hair density by intradermal administration of mesohyal NCTC 109 and mesohyal Biotin. mesoestetic Pharma Group S.L. 2016. **18.** Wound Healing Assay in primary dermal fibroblasts with X-DNA. Biotechnology Unit, Mesoestetic Pharma Group. 2016 Jun; INF89.UB.JUN.2016. **19.** Report on collagen synthesis – polynucleotides. Biotechnology Unit, Mesoestetic Pharma Group; INF105.UB.AGO.2016. **20.** Inhibition of gene expression of HYAL1 in fibroblasts irradiated with UVB. Biotechnology Unit, Mesoestetic Pharma Group; INF180.UB.MAR.2018.



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Scan the QR code to
display the principal
mesotherapy
injection techniques.



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